ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT

MATHEMATICS
Higher 1

8864 / 01

Paper 1

23 August 2012

JC 2 PRELIMINARY EXAMINATION

Time allowed**: 3 hours**

Additional Materials: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Index number, Form Class, graphic and/or scientific calculator model/s on the cover page.
Write your Index number and full name on all the work you hand in.
Write in dark blue or black pen on your answer scripts.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.
Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphic calculator are not allowed in the question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

This document consists of **6** printed pages.

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Anglo-Chinese Junior College

ANGLO-CHINESE JUNIOR COLLEGE
MATHEMATICS DEPARTMENT
JC2 Preliminary Examination 2012

MATHEMATICS 8864
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/ 95

Index No: ☐☐☐☐ Form Class: ___________

Name: ___________________________

Calculator model: ___________________

Arrange your answers in the same numerical order.

Place this cover sheet on top of them and tie them together with the string provided.

Question no.	Marks
1	/4
2	/5
3	/5
4	/3
5	/8
6	/10
7	/6
8	/8
9	/6
10	/6
11	/7
12	/13
13	/7
14	/7

Section A: Pure Mathematics [35 marks]

1 Without using a calculator, solve the inequality $x^2+2x-1 \le 3x+1$. **[2]**
Hence, using a sketch of the graph of $y=e^x$ or otherwise, solve the inequality $e^{2x}+2e^x-1 \le 3e^x+1$ exactly. **[2]**

2 Find

(i) $\int \frac{1}{2e^x}\,\mathrm{d}x$ **[2]**

(ii) $\int_0^k 2(x+k)^3\,\mathrm{d}x$ **[3]**

3 A curve C has equation $y=\frac{1}{2-x}$.

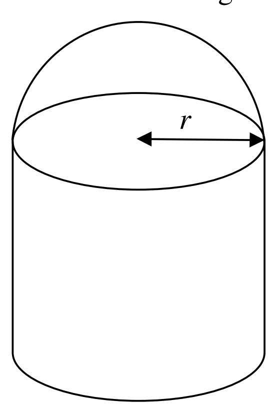
(i) Sketch C, stating the exact equations of any asymptotes and the exact coordinates of any intersections with the axes. **[2]**

(ii) Find $\int \frac{1}{2-x}\,\mathrm{d}x$. Hence find the exact area of the region bounded by C, the x-axis and the lines $x=-2$ and $x=1$. **[3]**

4 **(i)** Find the x-coordinates of the points of intersection of $y=0.1e^{2x}-1$ and $y=(3-0.5x)(5+x)$, giving your answers correct to 4 decimal places. **[1]**

(ii) Write down as an integral an expression for the area of the region bounded by $y=0.1e^{2x}-1$ and $y=(3-0.5x)(5+x)$. Evaluate this integral, giving your answer correct to 3 decimal places. **[2]**

5 The diagram below shows a cylindrical structure with a hemispherical roof top. The cost of each unit area of the roof is \$300 and the cost of each unit area of the curved surface area of the cylinder is \$100.
The total cost of the roof and the curved surface of the cylinder is \$3000. Given that the radius of the hemisphere is r units, show that the volume V of the structure can be expressed as $V = Ar - B\pi r^3$ where A and B are constants. **[4]**
Using a non-calculator method , find the exact cost of constructing the hemispherical top when the volume V is a maximum. **[4]**



[The formulae for
(a) Surface area and volume of a sphere are $A = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$ respectively.
(b) Curved surface area and volume of a cylinder are $A = 2\pi rh$ and $V = \pi r^2 h$ respectively.]

6 The equation of a curve C is $y = \ln(5 - 2x)$.

(i) Sketch the curve C, stating the exact equation of any asymptote and the exact coordinates of any intersection with the axes. **[2]**

(ii) Find $\dfrac{dy}{dx}$. **[1]**

(iii) The normal to the curve C at the point P where $x = -1$ meets the x-axis at Q and the y-axis at R. Find the exact coordinates of Q and R and hence show that the exact area of the triangle OQR is $\dfrac{1}{28}(a + b\ln 7)^2$ where a and b are integers to be found. **[7]**

Section B: Statistics [60 marks]

7 **(a)** An event organiser was tasked to find out from residents in a city suburb their choice vacation. In particular it is necessary to interview a representative range of ages.

(i) Explain how a quota sample might be carried out in this context. **[2]**

(ii) Explain a disadvantage of quota sampling in the context of your answer to part **(i)**. **[1]**

(b) The mean time spent by children watching television on Saturday is 3.5 hours with a standard deviation of 48 minutes. What is the probability that in a random sample of 100 children, the mean time spent watching television on Saturday is between 3.4 and 3.6 hours? **[3]**

8 For events S and T it is given that $P(S) = 0.4$, $P(T \mid S) = 0.7$ and $P(S' \cap T) = 0.3$.
Find

(i) $P(S \cap T)$ **[2]**

(ii) $P(T)$ **[2]**

(iii) $P(S \cup T')$ **[2]**

For a third event D, it is given that $P(D) = 0.25$ and that S and D are independent.
Find $P(S' \cap D)$. **[2]**

9 A box contains nine objects namely, two red and three white balls, two red and two white cubes.
Two objects are taken from the box, at random and without replacement.
Events B and R are defined as follows:

B : Both objects are balls,

R : Exactly one object drawn is red.

Find $P(B \mid R)$. **[3]**

If three objects are taken from the container, find the probability that at least one of each colour is obtained. **[3]**

10 A security firm transports a large sum of money between two places on a 5-day work week. The day on which the journey is made is varied at random.
Find the probability that, in a period of 10 weeks, Thursday will be chosen at most 3 times. **[2]**

In any 4-week period, show that the same day is chosen on all four occasions is given by 0.008. **[1]**

Find the probability that, in twenty 4-week period, the same day is chosen on all four occasions will occur more than once, giving your answer to 4 significant figures. **[3]**

11 Mrs T drives into town once a week to do her shopping. In the past, she parked her car at an allocated parking lot for $4.50, but she decided that in the next 5 weeks she would park at the roadside and risk getting tickets with fines of $10 per offence. The probability of getting a ticket each time she parks at the roadside is 0.15. What is the probability that Mrs T will pay more in fines in the 5 weeks than she would pay in parking fees if she had opted not to break the law? **[3]**

Taking a year to consist of 52 weeks, estimate the probability that Mrs T gets a tickets between 5 to 10 times(inclusive) in a year, using a suitable approximation. **[4]**

12 **(a)** A machine produces sheets of steel plates of thickness X cm where $X \sim \mathrm{N}(\mu, \sigma^2)$. Over a long period of time it has been found that 30% of the sheets are less than 29.5 cm thick and 85% of the sheets are more than 28 cm thick. Find to 2 decimal places, the value of σ. **[4]**

(b) The weekly takings, in dollars, at three cinemas are modelled as independent normally distributed random variables with means and standard deviations shown in the table below.

	Mean	Standard deviation
Cinema A	6000	400
Cinema B	5100	180
Cinema C	μ	σ

(i) Find the probability that in a randomly chosen week, the takings at cinema A will be less than the takings at cinema B. **[2]**

(ii) The probability that, in a randomly chosen week, the takings at cinema A exceeds x dollars is 0.975. Find x to the nearest dollar. **[2]**

(iii) The parent company receives a weekly levy consisting of 12% of the weekly takings at cinema A and 8% of those at cinema B. Find the probability that in a randomly chosen week, this levy exceeds $1150. **[3]**

(iv) Find the probability that exactly one week out of 5 randomly chosen week the takings of cinema C is less than μ. **[2]**

13 The breaking strength of a certain type of fishing line previously produced by a manufacturer was known to have breaking strength that is normally distributed with mean 8.75kN and standard deviation 0.24kN.

A new material is added to the fishing line being produced. The manufacturer believes that this will increase the breaking strength without changing the standard deviation. A random sample of 50 such new lines is tested and the mean breaking strength of the sample is $\bar{x}$ kN.

(i) Write down a suitable null and alternative hypothesis for the test in terms of μ. State the meaning of μ in this case. **[2]**

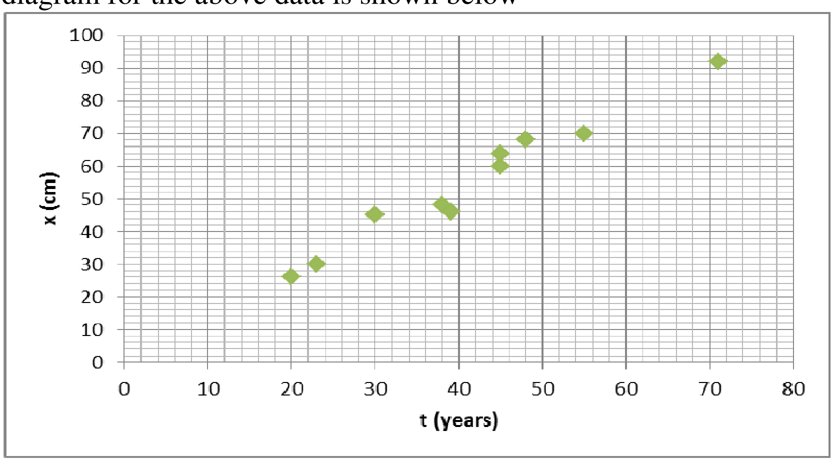
(ii) If $\bar{x} = 8.81$ kN, determine at the 10% significance level, whether there is sufficient evidence to conclude that the mean breaking strength has increased. **[3]**

(iii) Find the range of values of $\bar{x}$ for which the null hypothesis is rejected at the 5% significance level. **[2]**

14 A random sample of 10 oak trees in a timber yard was taken and girth of each trunk was measured. The following table shows the age and girth of each trunk .

Age (t years)	20	23	30	38	39	45	45	48	55	71
Girth (x cm)	26	30	45	48	46	60	64	68	70	92

The scatter diagram for the above data is shown below



(i) Calculate the product moment correlation coefficient between age and girth for these trees and comment briefly on your answer (with reference to the appearance of the scatter diagram) . **[2]**

(ii) Find the equation of an appropriate least squares regression line and use it to estimate the girth of a tree aged 60 years. **[2]**

(iii) It is suggested that the equation found in (ii) be used to predict the girth of a 100 years old tree. Comment briefly on the validity of this suggestion. **[1]**

(iv) From your equation find the predicted increase in girth per year of tree's age. **[1]**

(v) Under what circumstances would the least squares regression lines of girth on age and age on girth be coincident. **[1]**

- End of Paper -